

Dataset Personalization Methods based on LLMs for Data Science Education: A Comparative Study of Rescaling and Sampling Approaches

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Abstract

This work-in-progress study explores the use of Large Language Models (LLMs) to dynamically personalize datasets used in data science education according to learner interests. The study outlines two dataset personalization methods that transform each variable in the original dataset to a new variable: a scaling method that transforms while preserving the original distributions and correlation structure, and a sampling method that transforms while preserving the original correlation structure but changes distributions to match the new variables. An evaluation with subject matter experts revealed that datasets personalized using the scaling method were not significantly different from the original datasets in terms of the appropriateness of variable names and ranges. Further evaluation using the personalized datasets in the context of instructional materials designed for the original datasets indicated that more inconsistencies were found with the sampling method than the scaling method. These results suggest that dataset personalization can create datasets that serve as drop-in replacements for the original datasets in existing instructional materials.

CCS Concepts

• **Applied computing** → **Computer-assisted instruction.**

Keywords

Personalization, LLMs, Data Science Education

ACM Reference Format:

Luiz Barboza, Farshid Farzan, and Andrew M. Olney. 2025. Dataset Personalization Methods based on LLMs for Data Science Education: A Comparative Study of Rescaling and Sampling Approaches. In *Proceedings of the Twelfth ACM Conference on Learning @ Scale (L@S '25)*, July 21–23, 2025, Palermo, Italy. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3698205.3733936>

1 Introduction

Data science concerns fundamental principles that support and guide the principled extraction of information and knowledge from data [6] with a cycle including ingestion, storage and processing, analysis, and communication [4], has been dubbed the "sexiest

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L@S '25, Palermo, Italy

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ACM ISBN 979-8-4007-1291-3/2025/07

<https://doi.org/10.1145/3698205.3733936>

job of the 21st century" by Davenport et al. [1], and the field has seen a growing need for skilled data scientists, highlighted by a 35% projected job growth in the United States until 2032, according to the Bureau of Labor Statistics [11]. Despite the demand, learning data science poses challenges due to its evolving and interdisciplinary nature, as well as the prerequisite knowledge required, making it a complex area of study [8] that requires sustained study to master.

Context personalization of learning materials enhances engagement and helps students connect problems to their experiences, promoting deeper understanding and retention [2, 12]. By sparking situational interest and fostering a sense of connection and value in educational content, context personalization may foster a sustained individual interest in subjects like mathematics or computing [5]. Context personalization strategies can be based on preferences (e.g., friendships, food) or interests (e.g., music, sports) that reflect sustained engagement [3]. Recent research showed that Large Language Model (LLM) tools can offer a valuable opportunity to generate contextually and dynamically personalized learning materials, aligning educational content with learners' situational interests [5].

To address the motivational and engagement challenges in data science learning, this work-in-progress study explores the use of Large Language Models (LLMs) to dynamically personalize datasets based on learners' contextual interests. By integrating personalized datasets into data-driven examples and problem sets, we hope to leverage the known effects of context personalization for increasing motivation and fostering deeper learning. At this early stage of our research, our primary focus is whether the personalized datasets maintain the integrity of the original datasets, both independently and in the context of existing instructional materials. Our research questions are thus:

- To what extent can large language models be leveraged to generate contextually relevant and coherent variable names, ranges, and factor levels when personalizing diverse datasets across a range of topics?
- Is it sufficient for transformed variables to have plausible names and ranges, or is it necessary for transformed variables to have plausible distributions?
- How consistent are transformed datasets with instructional contexts designed around the original datasets?

2 Dataset Personalization Methods

We begin with an example to ground the problem of dataset personalization. Table 1 presents a sample from the iris dataset, which is perhaps the most widely used dataset in statistics education [10]. Datasets like the iris dataset are frequently used as examples

Table 1: The Iris Dataset

Sepal-L	Sepal-W	Petal-L	Petal-W	Species
5.1	3.5	1.4	0.2	setosa
7.0	3.2	4.7	1.4	versicolor
6.3	3.3	6.0	2.5	virginica

because they are small and simple, and instructors are familiar with their properties and what kinds of statistical phenomena they can be used to demonstrate. For example, the iris dataset has five variables that provide information about flowers, four of which are width or length measurements of the parts of an iris, and the fifth variable is species. This allows the iris dataset to be useful for teaching descriptive statistics, plotting, and even some advanced modeling topics.

However, irises may not be universally interesting to students, and further students may not be familiar the terminology of sepal, setosa, etc., which could make it more difficult for them to interpret results with this dataset. A naive approach to personalization would be to ask an LLM to generate a new dataset on a topic, e.g. baseball. However, that dataset would likely have little relationship to the original dataset and so not be useful in the same instructional contexts. We define the problem of dataset personalization as *transforming* an original dataset to a new dataset on a given topic, such that the new dataset can be used interchangeably with the original dataset in similar contexts.

A personalized version of the iris dataset must therefore have the same number of rows and the same number of columns as the original dataset. Additionally, the columns must be of the same type (4 numeric and 1 categorical) and categorical variables must have the same number of levels (here 3). Finally, the personalized dataset must have the same correlation structure as the original dataset, otherwise plots of one variable against another will have a different relationships between the original and personalized datasets.

With these criteria in mind, we introduce two approaches to dataset personalization, the scaling method and the sampling method. The scaling method transforms the original dataset while preserving the original distributions and correlation structure, while the sampling method transforms the original dataset while preserving the original correlation structure but changes variable distributions to match the real-world distributions of the new variables.

2.1 The Scaling Method

The scaling method begins by connecting to an LLM to generate a mapping from the original dataset schema to a new schema. The LLM is prompted with the original dataset columns and is asked to suggest new variables related to the target topic (see Appendix on OSF for prompts and code). Prompt instructions ask to ensure that the new schema matches the structure of the original dataset, maintaining the same column count and consistent categorical levels. The LLM then outputs a Python dictionary that maps each original column to its transformed counterpart, including a flag indicating if the variable is categorical, the new column name, and either a range for transformed numerical variables or levels for transformed categorical variables.

The scaling method then uses the information provided by the LLM to transform the original dataset. For numerical variables, the original data is scaled to fit the specified new ranges, which by definition preserves their relative distribution. For categorical variables, a dictionary-based replacement maps the original categorical levels to the new levels in a one-to-one mapping.

After the transformation, the personalized dataset contains new column names representing the target schema, with values rescaled or remapped to align with the new schema. For example, an original car dataset schema containing *mpg*, *cylinders*, *displacement*, *horsepower*, *weight*, *acceleration*, *modelyear*, *origin*, and *name* can be transformed into a target emissions dataset schema containing, *emissions*, *enginesize*, *fuelconsumption*, *poweroutput*, *vehiclemass*, *torque*, *productionyear*, *region*, *vehicletype*. The *mpg* column, for instance, could be rescaled to ‘emissions’ within a range of 50 to 400, while the *origin* column could be mapped to new regions such as North America, Europe, and Asia.

The scaling approach ensures that the new dataset maintains the correlation and distribution structure of the original data while aligning with the new thematic requirements of the target topic. A potential weakness of the scaling method is that distributions of the personalized dataset may not be realistic. For example *cylinders* is approximately bimodal, peaking a 4 cylinders and 6/8 cylinders. When mapped to *enginesize* using this method, it will similarly be bimodal, but the reality is that car engine sizes typically range from 1 to 3 liters and do not have a bimodal distribution. There is no restriction that precludes this type of mismatch from happening, though the LLM may implicitly avoid such transformations.

2.2 The Sampling Method

The sampling method attempts to solve the main disadvantage of the scaling method, which is that the distributions of transformed variables may not be realistic. While the motivational consequences of unrealistic transformations are not known, we suspect that any transformation perceived as unrealistic will be less motivating to the student.

The sampling approach avoids unrealistic distributions by using LLMs to predict the distribution of each new variable along with the minimum, median, and maximum values for each new variable. These values and distribution are then input to the *phitter* Python library, which predicts the best-fitting distribution parameters given these values. Once these distributions are known, we sample them, bounding by the minimum and maximum values using rejection sampling. The next step of the sampling method is to transfer the correlation structure of the original dataset to the new dataset. We match the correlation structure using a Gaussian copula constructed by the method of inversion [7, 9], which in brief requires transforming each variable to a uniform distribution using the probability integral transform. The uniform-transformed data is then multiplied by the Cholesky decomposition of the source dataset correlation matrix to transfer the correlation structure to the new data. Finally, the new dataset is back transformed using the inverse probability integral transform. Effectively this process shuffles the values in each column jointly such that the dataset overall has a correlation structure similar to the original. For discrete distributions, the LLM is additionally queried to rank order

levels for one of the continuous variables, e.g. in baseball the RBI of positions might be ordered Catcher, Center Fielder, Designated Hitter. This has no effect on the correlation structure but helps ensure that the discrete levels are realistically matched to other variables.

While the sampling method is more sophisticated than the scaling method, it is not clear that it is superior in practice. The sampling method makes several assumptions, including that the most important dependency structures are adequately represented by correlation, i.e. are linear, and that the Gaussian copula can adequately capture correlation structure across variables of mixed types (an empirical assumption). These assumptions, together with the increased technical complexity of the sampling method entails that there are more opportunities for error than the scaling method.

The sampling method effectively trades some fidelity in the correlation structure for increased fidelity in distributions, and it is not currently known which is more important for effective instruction. Finally, it should be noted that neither the scaling nor sampling method attempt to adjust the realism of the correlation matrix for the transformed data, which in both methods is determined by the source data. This decision is due to the technical problem that both methods attempt to address: how can we adapt instruction around existing datasets to the personal interests of learners without significantly changing the instructional material using the original dataset?

3 Method

3.1 Design

The study used a repeated measures design in which three subject matter experts (SMEs) who were blind to the study goals rated original datasets, scaling-transformed datasets, and sampling-transformed datasets. Ratings were either on the dataset out of context or the dataset in the context of a Jupyter notebook teaching a data science concept. The primary outcome measures of the study were the absolute ratings across datasets, both in and out of context. Because the personalization transformation creates multiple personalized datasets for every original dataset, this design has more power for detecting differences between scaling and sampling methods than between these methods and the original datasets. Data were analyzed using beta regression for comparison of all three groups and Wilcoxon signed-rank tests for comparison of scaling and sampling methods. Beta regression is appropriate for continuous bounded outcome variables, unlike linear regression, which isn't suitable for bounded outcomes, and logistic regression, which can be used for proportions, but only when the proportion is a ratio of two counts [24]. Because beta regression is defined on the open interval (0,1), we use a standard transformation to squeeze our closed interval outcome variables to the open interval [25].

3.2 Materials

The original datasets included the Iris and Breast Cancer Wisconsin (Diagnostic) datasets from the UCI Machine Learning Repository as well as the Boston and Baseball datasets from the StatLib library at CMU. Target topics were selected based on Google Trends' most popular topics to ensure that the topics were unbiased, diverse, and

representative of popular topics that might be used in personalization: Sports, Travel, Games, and Health. The four original datasets were transformed into personalized datasets on the four topics using the scaling method and sampling method, both using GPT4 (gpt-4-turbo-2024-04-09), to generate 32 personalized datasets, 16 for each method. Together with the original datasets, there were a total of 36 datasets for evaluation.

Two sets of materials were created for SMEs. The out-of-context evaluation materials consisted of a spreadsheet with columns for topic, variable name, and variable range (or levels for a categorical variable). Each row contained two questions, "The variable name matches the topic" and "The range of the variable is plausible based on your knowledge," each of which could be answered on a six point Likert scale, ranging from Strongly Agree to Strongly Disagree. The in-context evaluation materials consisted of a spreadsheet and a set of four Jupyter notebooks. The spreadsheet contained columns for notebook, topic, and method (0, 1, or 2), which combined identify an evaluation condition for the Jupyter notebook. Each row contained one question, "Count the number of problems in the notebook," with multiple examples of problems were given, e.g. notebook text referring to linear/nonlinear patterns in plots that aren't present. The Jupyter notebooks covered four methods: descriptive statistics, KNN regression, regression trees, and random forests. These notebooks were originally written for the Iris, Boston, Baseball, and Breast Cancer datasets, respectively.

3.3 Procedure

The researchers met with the three SMEs to explain the ratings tasks. For the out-of-context ratings, SMEs evaluated each row (a variable in a dataset) and gave a rating for variable name and variable range on the six-point Likert scale using a dropdown to prevent invalid data entry. SMEs completed 645 variable-level, out-of-context ratings. For the in-context ratings, SMEs used the notebook, topic, and method identifiers in the spreadsheet to configure Jupyter notebooks to load the correct dataset. Thus there were only four Jupyter notebooks, one per original dataset, and these four notebooks were configurable by SMEs to use the 32 personalized datasets as well as the 4 original datasets. SMEs loaded each dataset in the notebook using dropdowns to prevent invalid data entry, ran the cells in the notebook, and then examined the notebook for any problems. SMEs then entered a whole number in the spreadsheet indicating their count of the problems in the notebook.

Table 2: Ratings across SMEs

SME	n	Name		Range		n	Problems	
		M	SD	M	SD		M	SD
1	645	5.99	.15	5.93	.47	36	2.06	.83
2	645	5.99	.16	5.95	.45	36	2.17	.88
3	645	5.78	.57	5.92	.33	36	1.25	.60

4 Results

Descriptive statistics of the SME's ratings are shown in Table 2. All SMEs tended to give high scores for the name-match and plausible-range questions. Further inspection of the distributions of these

ratings indicated that SMEs 1 and 2 tended towards binary ratings of these questions with “Strongly agree” being dominant, whereas SME 3 gave slightly more graduated ratings while still primarily endorsing “Strongly agree.” A similar trend was seen for the count of problems: inspection of the distributions revealed that ratings for SMEs 1 and 2 were similarly distributed, and ratings for SME 3 were comparable but shifted downward, such that SME 3 tended to see fewer problems. Because the ratings from all 3 SMEs were so similar, we averaged their ratings for all further analyses.

Table 3: Ratings across methods

Measure	Scaling			Sampling		Original		
	n	M	SD	M	SD	n	M	SD
Name	276	5.93	.19	5.92	.19	69	5.94	.14
Range	276	5.96	.15	5.92	.37	69	5.86	.29
Problems	16	1.62	.64	2.10	.42	4	1.50	.96

Descriptive statistics using combined ratings for SMEs are shown in Table 3 for the scaling- and sampling-personalized datasets as well as the original datasets. While the original datasets have the highest average scores and lowest problems, the scaling and sampling methods are quite close.

To answer our first research question of how well LLMs generate variable names matching a given topic and plausible ranges for those variables, we ran beta regressions to predict these outcomes using method (original, scaling, or sampling) as the only predictor. Data used in the regression were aggregated by source dataset, method, and variable index, which had no effect on the original datasets but averaged each variable for the scaling and sampling methods, e.g. for a given dataset and method, measures for the variable in the first column were averaged to create a single score for that column ($n = 69$). Measures were further transformed to Beta regression for ratings of whether the variable names match the topic showed a significant effect of method $\chi(2) = 7.59, p = .022$. Pairwise contrasts using Tukey’s test revealed that variable names match topics in the original dataset ($M = .989, SE = .001$) more than the sampling method ($M = .984, SE = .002$), $p = .021$, but there was no difference between the original method and the scaling method ($M = .986, SE = .002$), $p = .143$, or between the scaling and sampling methods, $p = .695$. Beta regression for ratings of whether the variable ranges were plausible showed no effect of method $\chi(2) = .499, p = .779$.

To answer our second research question of whether plausible distributions were important in these ratings, i.e. whether the scaling method is more positively rated than the sampling method, we conducted two Wilcoxon signed-rank tests using de-aggregated data ($n = 276$), which gives us the maximum amount of power to detect an effect allowed by our design. The signed-rank test for ratings of variable names matching the topics found no significant difference between the scaling and sampling methods, $V = 865, p = .942$, and likewise the signed-rank test for ratings of variable ranges being plausible found no significant difference between the scaling and sampling methods, $V = 429, p = .390$.

To answer our third research question of whether the transformed datasets are consistent with instructional contexts designed

around the original datasets, we conducted a Wilcoxon signed-rank test for problems found in notebooks ($n = 32$). The signed-rank test for problems found in notebooks found a significant difference between the scaling and sampling methods, $V = 20.5, p = .048$, such that more problems were found in sampling notebooks ($M = 2.10, SD = .42$) than in scaling notebooks ($M = 1.62, SD = .64$). We do not report results of a beta regression comparing all three conditions because aggregated data would have only 4 observations per condition (i.e., there are only 4 original notebooks). However, we note that problems in the original notebooks ($M = 1.50, SD = .96$) trended lower than the scaling notebooks, suggesting that in a sufficiently powered design, a significant difference in problems might be found between the sampling method and the original notebooks.

5 Discussion and Future Work

This study explored the use of Large Language Models (LLMs) to dynamically personalize datasets used in data science education according to learner interests. We introduced two dataset personalization methods, the scaling method and the sampling method, which preserve the key properties of the original dataset while transforming it to a new topic. Our results indicate that the scaling method is a more robust approach to data personalization than the sampling method, which was observed to have less plausible variable names than the original datasets and had more problems when substituted for the original datasets in Jupyter notebooks than did the scaling method. Since the two methods primarily differ on whether they preserve the distributions of the original variables, it is unclear why the sampling method would be less likely to produce variable names that match the target topic. It is possible that the additional information the sampling method included in the LLM prompt, necessary to generate new distributions, may have negatively impacted the LLM generation of variable names. However, our other main finding that sampling method datasets are less acceptable than scaling method datasets as a drop-in replacement for the original dataset, is understandable since changing the variable distributions may have unforeseen consequences for existing instructional materials. Altogether, the sampling method is more complex and has more opportunities for error, which may make it more suitable (with current LLM technology) to be used in a semi-automated way, whereas the scaling method could be deployed for automated data personalization based on our results.

A limitation of our study is that we only evaluated transforming four datasets to four topics, so results may change with a larger number of topics or datasets. Additionally, our topics were very high level, e.g. Sports, and it is unclear if LLMs would perform similarly on this task for more specific topics, e.g. Taylor Swift’s Eras Tour. Future work should examine the effect of topic specificity on data personalization as well as the effects of data personalization on the learning process. Past work on context personalization indicates that data personalization could lead to enhanced motivation and learning outcomes, but whether this effect extends to data personalization (and to what extent) remains to be seen.

Acknowledgments

This material is based upon work supported by the National Science Foundation under Grant No. 1918751.

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